



Knowledge-driven quantum architecture search through filtering and focusing

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ABSTRACT

Quantum architecture search (QAS) represents a fundamental challenge in quantum machine learning. The success of various QAS raises a fundamental question about what enables efficient quantum architecture design and how to systematically analyze search performance. From a knowledge-driven perspective, we decompose QAS into a progressive search process with continuous knowledge accumulation and transfer. We propose a framework through knowledge-guided untrained filtering and trained focusing for iterative architecture discovery with knowledge refinement. The framework develops an evolutionary mechanism to enhance multi-level knowledge transfer in architecture searching. Experiments demonstrate improvements across diverse tasks: achieving 59.1% performance increase in signal representation, maintaining robust performance in both simulated and real-noise environments for image classification, and improving precision by 9 orders of magnitude in variational quantum eigensolver tasks, providing a viable approach for QAS.

1. Introduction

Quantum Machine Learning (QML) aims to reinterpret and enhance classical machine learning within the framework of quantum computing. QML not only inherits the wide applicability of traditional machine learning, encompassing diverse tasks from classification [1] and regression to generative modeling [2] and reinforcement learning [3], but also incorporates the unique advantages of quantum computing. In the era of Noisy Intermediate-Scale Quantum (NISQ) devices, QML has garnered significant attention for its potential to solve complex real-world problems.

Currently, Variational Quantum Circuits (VQCs) represent the most widely adopted method in QML [4]. VQCs optimize model performance through iterative optimization of quantum circuit parameters until the quantum circuit can learn and represent the input problem with high accuracy [5]. From this perspective, QML can be abstracted as an architecture optimization problem guided by a loss function and subject to multiple quantum constraints. While VQCs provide a powerful framework for quantum machine learn-

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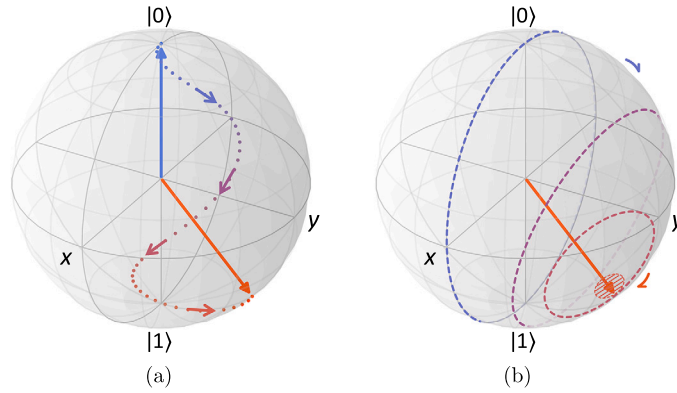


Fig. 1. Comparison between gradient-based and training-free QAS methods. Output single-qubit states are used to visualize the circuit search process. (a) Continuous gradient-based approach gradually converges to target circuit through gradient guidance. The arrows represent the gradient direction for the next search step from the current position. (b) Discrete training-free approach narrows search space through screening indicators to locate target circuit vicinity. The circles represent the search space during each iteration of the search process.

ing, designing efficient quantum circuits faces significant challenges as problem scales increase. Traditional manual design methods, although highly task-specific, often struggle to handle complex, large-scale problems effectively. Conversely, the adoption of generalized template-based quantum circuit designs, while simplifying the design process, frequently leads to performance bottlenecks, particularly when confronted with issues such as barren plateaus [6].

To tackle quantum circuit design, various QAS methods have been proposed. Similar to Neural Architecture Search (NAS), QAS can be viewed as an item retrieval task within a vast architecture space [7]. The core challenges of QAS as a retrieval problem are: (1) efficiently indexing, searching, and evaluating circuit quality in an exponential architecture space, (2) leveraging search history to improve retrieval accuracy while maintaining search diversity, and (3) balancing between search efficiency and precision under computational constraints. From a knowledge perspective, QAS can be reformulated as a knowledge-driven retrieval task in a vast architecture space with limited prior information, where the efficiency and effectiveness heavily rely on how to acquire and utilize knowledge during architecture search. This perspective enables us to unify existing QAS approaches as specialized retrieval methods leveraging different aspects of quantum circuit knowledge, while also revealing opportunities for more comprehensive knowledge utilization. Recent years have witnessed diverse approaches to QAS, each providing unique insights into quantum circuit design.

Heuristic algorithms explore quantum architectures through different predefined strategies: evolutionary methods [8,9] that enable population-based exploration, genetic programming methods [10] adapt evolutionary principles for circuit optimization, quantum autoencoder approaches [11] leverage neural network principles for resource optimization, and metaheuristic-based methods [12] systematically explore the architecture space through similarity-guided search. While these approaches provide strong interpretability through their explicit search rules, they often face efficiency challenges in large-scale search spaces.

Differential-based QAS methods have evolved to enable end-to-end optimization: gradient-based frameworks [13,14] construct differentiable search spaces for direct optimization, Gumbel-Softmax based approaches [15] enable efficient architecture sampling, neural predictor guided methods [16] accelerate architecture evaluation, and continuous evolution strategies [17] address the challenges posed by the worse search space. These approaches effectively leverage optimization knowledge but are often constrained by their preset search spaces and differentiability requirements.

Training-free QAS represents a recent trend focusing on rapid architecture screening: noise-guided frameworks [1] evaluate circuits based on hardware constraints, while proxy-based methods [7] utilize circuit characteristics like expressibility for efficient filtering. Though achieving remarkable search efficiency, these approaches may occasionally overlook promising architectures due to their simplified evaluation metrics.

With recent advances in machine learning methodologies, pioneering works have begun to leverage sophisticated learning models to automatically discover and exploit key features in the QAS process. The field has further expanded with the introduction of reinforcement learning-based approaches [18] that dynamically accumulate design experience, reinforcement learning guided exploration [19] iteratively refines architectures based on noise-aware feedback, and representation learning techniques [20] that capture latent circuit patterns. Recent transformer-based methods [21] have demonstrated the potential of attention mechanisms in understanding circuit topology. These learning-based approaches represent the cutting edge of QAS research, offering deeper insights into quantum circuit characteristics and enabling more sophisticated understanding of architecture design principles. See Fig. 1.

The evolution and success of these diverse approaches reveal a fundamental insight: each method essentially discovers and utilizes different aspects of quantum circuit knowledge - from structural patterns and optimization dynamics to representation capabilities - albeit typically focusing on specific aspects within their respective paradigms. This observation suggests an opportunity for a more comprehensive knowledge integration framework in QAS that combines the strengths of different knowledge types. These approaches can be analyzed through their implicit knowledge utilization patterns, revealing different strategies for quantum circuit understanding. In the context of QAS, knowledge represents the guiding principles and learned patterns that inform architecture retrieval decisions. Different types of knowledge guide the retrieval process by providing search priorities and evaluation criteria. The effectiveness of QAS methods largely depends on how they leverage such knowledge to guide the retrieval of optimal architectures. This observation

suggests an opportunity: rather than treating QAS as separate methodological tracks, we can view it as a knowledge-driven process where different types of circuit understanding guide the search. This perspective reveals the potential for more comprehensive knowledge utilization in quantum architecture design. From this knowledge-driven perspective, we can categorize quantum circuit understanding into four complementary types: structural, expressivity, optimization, and task-specific knowledge, each capturing different aspects of circuit properties.

Building upon these advances, we propose **Quantum Untrained-Explored Synergistic Trained Architecture (QUEST-A)**, a framework that integrates and transfers multiple types of quantum circuit knowledge throughout the search process. QUEST-A achieves this through three coupled components: (1) knowledge-guided untrained filtering for search space reduction through topological analysis and expressivity assessment, (2) knowledge-accumulative trained focusing for precise architecture refinement through parameter optimization and reuse, and (3) evolutionary synthesis for continuous knowledge transfer across generations. This integrated approach enables progressive knowledge accumulation and utilization, addressing the challenges in space exploration, knowledge transfer, and search balance.

The contributions of our work are as follows:

1. Formulated QAS as a knowledge-driven search problem and developed a framework that simultaneously addresses three fundamental challenges: search space efficiency through knowledge-guided filtering, knowledge accumulation continuity through evolutionary mechanisms, and exploration-exploitation balance through multi-level refinement.
2. Developed an evolutionary mechanism with reupload circuits and dual-evaluation strategies, integrating coarse-grained untrained filtering and fine-grained trained focusing through knowledge accumulation and reuse, which provides a technical solution for progressive optimization.
3. Demonstrated improvements across diverse tasks through comprehensive experiments: enhancing model expressivity by up to 59.1% in signal representation, maintaining robust performance in both simulated and noisy environments for image classification, and improving precision by 9 orders of magnitude in VQE tasks.

2. Related works

2.1. Ansatz design for quantum machine learning

VQCs have emerged as the dominant methodology in QML. A quantum circuit ansatz, serving as the fundamental template for QML models, comprises modular components for data encoding, trainable variational gates, and measurement operations, thereby forming a quantum-classical hybrid computational process. In addition to traditional cascading CNOT circuits [6], the advancement in understanding quantum circuit characteristics [22] has given rise to various novel ansatzes. Re-uploading circuits [23] alternate data encoding and parameter variation processes, enabling multiple utilizations of input data and thus enhancing the circuit's expressivity. This design establishes a complex, data-oriented non-unitary process that can approximate any continuous function [24], while providing increased flexibility and controllability in the search space. More importantly, by incorporating data encoding into the optimization scope [25], re-uploading circuits' modular and repeatable structure provides an ideal template for efficient knowledge transfer and accumulation during architecture searching.

2.2. Formulation of quantum architecture search

Quantum Architecture Search can be formalized as an optimization problem in a large-scale discrete search space [13]. For a quantum model comprising n qubits and m layers, where each layer can select from a set of quantum gates G with cardinality $k = |G|$, the search space size reaches $O(k^{n \times m})$ [26]. While its exact computational complexity remains open, QAS is conjectured to be at least NP-hard [15]. Let the candidate quantum gates be represented by matrix $M \in G^{n \times m}$, define $U_{ij} = \sigma(M_{ij})$, where σ maps the quantum gate on the i -th qubit and j -th layer to a $2^n \times 2^n$ unitary matrix. The unitary matrix corresponding to the entire quantum circuit can be expressed as:

$$U = \bigotimes_{j=1}^m \bigotimes_{i=1}^n U_{ij} = \bigotimes_{j=1}^m \bigotimes_{i=1}^n \sigma(M_{ij}) \quad (1)$$

further, the search objective can be formalized as $\min_{M \in G^{n \times m}} L(U, D)$, where L denotes the loss function, and D represents the dataset or target unitary matrix for the specific problem. The core of QAS lies in efficiently retrieving the quantum circuit structure M and its parameters that minimize the corresponding loss function within the $O(k^{n \times m})$ -scale search space with limited prior knowledge. This exponential complexity indicates that the effectiveness of QAS heavily relies on how to acquire and utilize knowledge during the search process. The selection of high-quality quantum circuit templates as starting points becomes crucial for reducing the search space size k and making the $O(k^{n \times m})$ space more tractable. Subsequently, effective knowledge acquisition and transfer mechanisms are essential for navigating this vast search space efficiently.

3. Formulation and methodology

3.1. Knowledge-driven formulation of QAS

As defined earlier, knowledge in QAS represents the guiding principles that inform architecture retrieval decisions through search priorities and evaluation criteria. In this work, we propose to reformulate QAS from a knowledge-driven perspective [27], where the search process is decomposed into the acquisition, accumulation, and utilization of different types of knowledge about quantum circuits. This reformulation not only provides a systematic way to understand and analyze QAS but also suggests effective strategies for navigating the vast architecture space through progressive knowledge refinement.

To formalize this knowledge-driven perspective, we first establish the mathematical framework for representing and analyzing knowledge in QAS. Given a quantum circuit search space $S \subseteq \mathcal{G}^{n \times m}$, we define a knowledge state $\mathcal{K}$ as a tuple

$$\mathcal{K} = (S, C, \pi, \mathcal{M}_m), \quad (2)$$

where S represents the candidate architecture space following formulation in Section 2.2, $C \subseteq S$ denotes the currently explored subspace containing architectures that have been evaluated, $\pi : S \rightarrow [0, 1]$ is a possibility distribution over architectures reflecting our understanding of their potential, and $\mathcal{M}_m$ represents the set of observed measurements including both circuit properties and performance metrics. This formulation enables us to systematically track and analyze the knowledge accumulated during the search process. Based on this framework, we categorize the knowledge in QAS into *four* complementary types, which together provide a complete characterization of quantum circuits from different perspectives.

The first type is structural knowledge $\mathcal{K}_s$, which captures the topological and compositional properties of quantum circuits:

$$\mathcal{K}_s = \{(f_s(M), g_s(M)) | M \in S\}, \quad (3)$$

where $f_s(\cdot)$ represents structural complexity measures such as circuit depth or CNOT gate count and $g_s(\cdot)$ denotes architectural features [28,29] like gate connectivity patterns or overall structures. This knowledge is essential for understanding the fundamental properties of circuit architectures and their potential computational capabilities.

The second type is expressivity knowledge $\mathcal{K}_e$, which characterizes the quantum state representation capability:

$$\mathcal{K}_e = \{(f_e(U), g_e(U)) | U = \prod \sigma(M), M \in S\}, \quad (4)$$

where $f_e(\cdot)$ measures the expressivity of quantum states and $g_e(\cdot)$ describes their distribution characteristics, both of which together exhibit the reachability of variational quantum circuits in Hilbert space. The unitary matrix U is constructed through σ mapping as defined in Section 2.2. This knowledge directly reflects a circuit's ability to represent complex quantum states.

The third type is optimization knowledge $\mathcal{K}_o$, which encodes the information about parameter optimization:

$$\mathcal{K}_o = \{(\theta, \nabla L(\theta), \eta(\theta)) | \theta \text{ parameters of circuits in } C\}, \quad (5)$$

where $\nabla L(\theta)$ represents the gradient information and $\eta(\theta)$ indicates the optimization trajectory. This knowledge is particularly crucial in QAS due to the intrinsic coupling between circuit architectures and their parameters.

The fourth type is task-specific knowledge $\mathcal{K}_t$, which represents the actual performance measurements:

$$\mathcal{K}_t = \{(M, L(U, D), v(M)) | M \in C\} \quad (6)$$

where $L(U, D)$ is the task-specific loss function and $v(M)$ captures additional performance characteristics. Together, they reflect the specific performance of variational quantum circuits when applied to specific task scenarios.

The completeness of this knowledge categorization is supported by the following relationship:

$$\forall M \in S : \text{Quality}(M) = F(\mathcal{K}_s(M), \mathcal{K}_e(M), \mathcal{K}_o(M), \mathcal{K}_t(M)), \quad (7)$$

where F represents the functional mapping from different types of knowledge to the overall quality of a circuit. This relationship demonstrates that these four types of knowledge collectively determine and fully characterize the properties and performance of quantum circuits in the context of QAS. The effectiveness of this knowledge categorization in QAS is supported by theoretical analysis in Appendix B, where we establish the complexity bounds of knowledge-guided search and prove the convergence of this knowledge-driven process.

Guided by these theoretical insights, we propose QUEST-A, a framework which realizes QAS driven by four kinds of knowledge through three coupled components.

3.2. Overview of QUEST-A

As shown in Fig. 2, QUEST-A consists of three core components for progressive knowledge accumulation and utilization:

1. **Knowledge-Guided Untrained Filtering:** Guided by structural knowledge and expressivity knowledge, this component employs DAG-based analysis and KL-divergence estimation to efficiently reduce the effective search space. This component primarily utilizes structural knowledge ($\mathcal{K}_s$) and expressivity knowledge ($\mathcal{K}_e$) for efficient search space reduction.

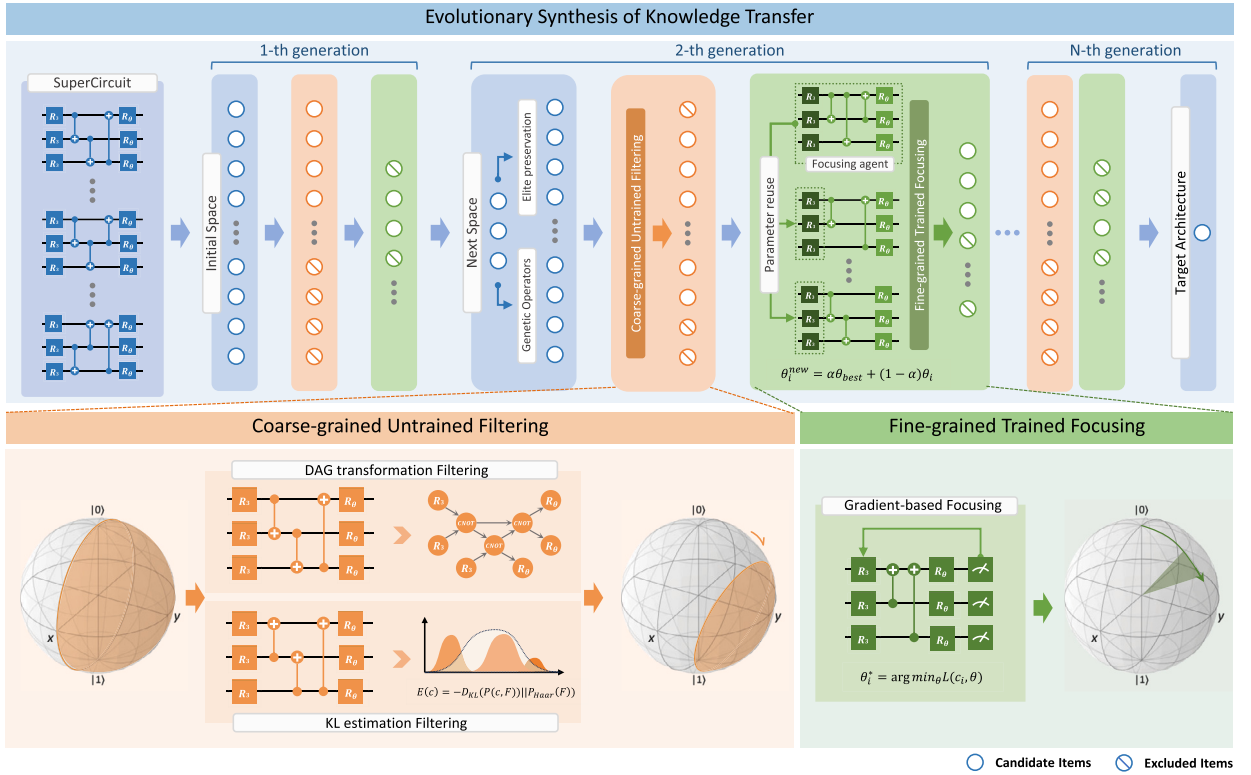


Fig. 2. QUEST-A framework.

2. **Knowledge-Accumulative Trained Focusing:** Utilizing optimization knowledge and task-specific knowledge, this component implements gradient-guided optimization with parameter reuse strategies for precise architecture refinement. This component leverages optimization knowledge ($\mathcal{K}_o$) and task-specific knowledge ($\mathcal{K}_t$) for precise architecture refinement.
3. **Evolutionary Synthesis of Knowledge Transfer:** This component provides a unified framework for knowledge accumulation and transfer through evolutionary operations, ensuring efficient knowledge preservation and inheritance between search generations. This component integrates and transfers all four knowledge types across generations, ensuring comprehensive knowledge inheritance.

These three coupled components form a closed-loop search system that implements our knowledge-driven approach.

3.3. Untrained filtering

Given the number of qubits N , maximum circuit depth L , and a set of available quantum gates $G(|G| = Q)$, we adopt a dynamic indexing strategy where the indexer A for the search space S is defined as $A : \{0, 1, \dots, Q-1\}^{N \times L} \rightarrow S$, where $\{0, 1, \dots, Q-1\}^{N \times L}$ represents the $N \times L$ -dimensional discrete index space. In the i -th iteration, the sampled specific quantum circuit structure a^i can be expressed as $a^i = A(i^i)$, $i^i \in \{0, 1, \dots, Q-1\}^{N \times L}$. This dynamic indexing method enhances memory efficiency by avoiding the need to store the entire supercircuit structure [19,30], while also improving computational efficiency by generating and evaluating specific structures only when necessary.

Furthermore, through the integration of re-uploading circuits in Section 2.1, QUEST-A enables the search for optimal data embedding methods [23]. This design naturally supports the accumulation of structural knowledge through modular components and facilitates expressivity knowledge transfer through repeated data processing patterns.

Building upon the QUEST-A framework, the intrinsic properties of quantum circuits are further utilized to optimize the search process. These properties, directly related to the structural complexity and potential expressivity of the circuits, enable rapid identification and exclusion of candidate circuits with evident deficiencies in circuit scale and expressivity, thereby narrowing the search space [7]. QUEST-A employs two complementary evaluation methods: topological analysis based on Directed Acyclic Graphs (DAGs) and expressivity assessment using Kullback-Leibler (KL) divergence.

To effectively utilize structural knowledge, quantum circuits are represented as DAGs [28], where nodes represent quantum gate operations and edges represent qubit evolution paths. Based on this representation, the number of paths from input to output in the DAG is used as a proxy indicator for circuit complexity. For a given quantum circuit c , its path count is defined as $P(c)$. As

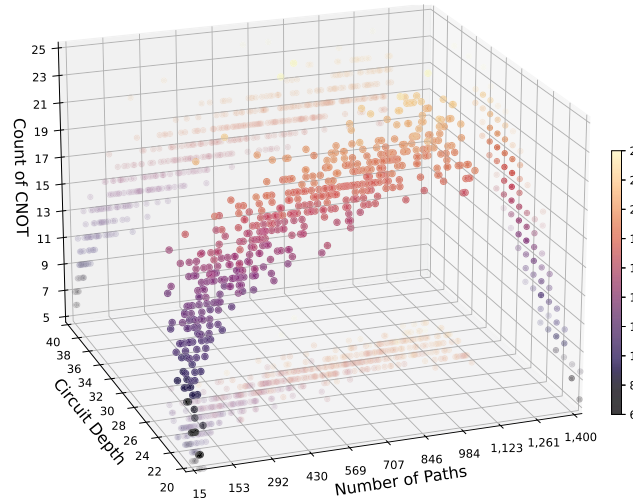


Fig. 3. Relationship between DAG-path count and circuit complexity for 500 random quantum circuits. As the DAG-path count increases, both circuit depth and number of CNOT show logarithmic growth, indicating a strong correlation between DAG-path count and circuit scale.

demonstrated by the empirical study in Fig. 3, $P(c)$ generally exhibits a positive correlation with the number of CNOT gates and circuit depth. This correlation highlights the effectiveness of $P(c)$ as an indicator of circuit complexity.

The use of $P(c)$ aims to balance between circuit expressivity and trainability, selecting circuits that potentially avoid both under-parameterization and the barren plateau problem associated with over-parameterization, as suggested by studies [28,29] on quantum circuit complexity and trainability. This screening mechanism effectively eliminates candidate circuits that clearly do not meet task requirements.

To evaluate expressivity knowledge, we introduce an assessment $D(c)$ based on KL divergence, which quantifies circuit expressivity by comparing the quantum state distribution with the ideal *Haar* random distribution. This method quantifies circuit expressivity by comparing the quantum state distribution generated by the circuit with the ideal Haar random distribution [31]. Specifically, for a quantum circuit c , its expressivity can be quantified by the following:

$$E(c) = -D_{KL}(P(c, F) || P_{Haar}(F)), \quad (8)$$

where $P(c, F)$ is the fidelity distribution of quantum states generated by circuit c , $P_{Haar}(F)$ is the Haar random distribution, and D_{KL} denotes KL divergence. $F = |\langle \psi_\theta | \psi_\phi \rangle|^2$ defines the fidelity of quantum states, where θ and ϕ are the parameters of circuit c . While the exact value of quantum state fidelity can be calculated from the density matrices before and after quantum state evolution, in practice, $P(c, F)$ is approximated by random parameter sampling of circuit c . The analytical form of $P_{Haar}(F)$ is:

$$P_{Haar}(F) = (N-1)(1-F)^{N-2}, \quad (9)$$

where N is the dimension of the Hilbert space. A larger $E(c)$ value indicates higher expressivity, reflecting the circuit's ability to explore a larger portion of the Hilbert space.

Topological analysis, though effective for circuit complexity evaluation, cannot capture quantum-specific properties of circuits. KL divergence complements this limitation by providing a quantitative measure of circuit expressivity. This approach aligns with work research [32] on quantum circuit expressivity, emphasizing the importance of state distribution uniformity in evaluating circuit performance. In practical work, the relationship between KL divergence and candidate circuit performance does not exhibit a simple linear correlation. Instead, it provides a range of circuits with relatively stable performance, which can be used to filter out a series of circuit structures with lower training potential. See Fig. 4.

By combining these two knowledge-guided evaluation methods, we achieve search space reduction as demonstrated in *Theorem 1*. This filtering process directly operationalizes structural knowledge through DAG-based path counts and expressivity knowledge through KL divergence assessment, effectively translating theoretical knowledge types into practical filtering mechanisms.

3.4. Trained focusing

For optimizing circuits in the reduced search space guided by Theorem 1, given a set of candidate quantum circuits $\mathcal{P}'_g = \{c_1, c_2, \dots, c_K\}$, where each circuit c_i is composed of a structure A_i and parameters θ_i , we utilize gradient-guided optimization to accumulate optimization knowledge and task-specific knowledge through

$$\theta_i^* = \arg \min_{\theta} L(c_i, \theta), \quad (10)$$

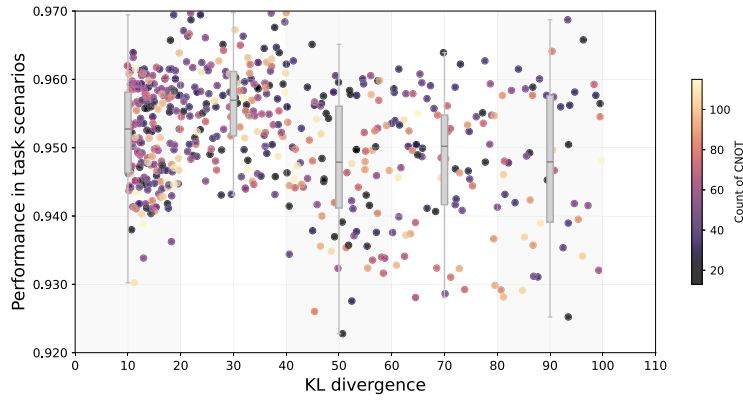


Fig. 4. Relationship between KL divergence and quantum circuit performance in task scenarios. The test evaluates KL divergence for 500 random circuits using data re-uploading circuits as templates. These circuits were trained on an MNIST 4-class classification task until convergence. Outliers have been removed.

where $\mathcal{L}(c_i, \theta)$ is the loss function for circuit c_i . However, as the number of candidate quantum circuits increases, the computational resources required for gradient descent optimization grow exponentially.

To ensure efficient knowledge transfer as established in Theorem 2, we develop a parameter reuse strategy that enables knowledge accumulation across the optimization process. Within the evolutionary algorithm framework of QUEST-A, parameter reuse serves as an effective mechanism for knowledge transfer and accumulation across generations. Specifically, in each generation of candidate circuits, the best-performing individual is designated as the focusing agent. After fine-grained trained focusing, its parameters are selectively reused by other individuals in the next generation, thus disseminating valuable knowledge throughout the entire population.

Specifically, let c_{best} be the best-performing individual serving as the focusing agent

$$c_{best} = \arg \max_{c \in P'_g} F(c), \quad (11)$$

where $F(c)$ is the fitness function incorporating both structural and expressivity assessments

$$Fitness(c) = f(P(c), E(c)). \quad (12)$$

The parameter reuse strategy is formalized as

$$\theta_i^{new} = \alpha \theta_{best} + (1 - \alpha) \theta_i \quad (13)$$

where $\alpha \in [0, 1]$ is the reuse rate, and θ_{best} are the parameters of c_{best} . In actual tasks, α is set to be very close to 1 to balance the utilization of posteriori knowledge and the diversity of search.

The parameter reuse strategy represents a concrete implementation of optimization knowledge transfer, while the fitness function integrates task-specific knowledge to guide precise architecture refinement. This parameter reuse strategy, combined with the knowledge-guided filtering, ensures continuous knowledge accumulation during optimization as shown in Theorem2, while maintaining the convergence rate established in Theorem3.

3.5. Evolutionary synthesis of knowledge transfer

To systematically implement the knowledge transfer mechanism established in our theoretical framework, we integrate the knowledge-guided filtering and knowledge-accumulative focusing into a unified evolutionary framework. The evolutionary process naturally supports both coarse-grained and fine-grained knowledge accumulation, where population evolution corresponds to architecture space exploration and fitness evaluation reflects multi-level knowledge assessment. The evolutionary synthesis framework consists of four key modules that interact sequentially: (1) population initialization and sampling, (2) knowledge-guided evaluation using both untrained filtering and trained focusing, (3) knowledge transfer through evolutionary operations, and (4) final selection. Each generation progressively refines the architecture space by evaluating candidates against multi-dimensional knowledge criteria and preserving beneficial architectural patterns. The framework of evolutionary synthesis is described in Algorithm 1.

This evolutionary framework provides a unified mechanism for implementing the knowledge transfer established in Theorem 2, where the fitness evaluation combines structural and expressivity knowledge, while the evolutionary operations ensure efficient knowledge inheritance between generations. Specifically, the update of superstructure S preserves structural and expressivity knowledge, the evolution operations through crossover and mutation facilitate architectural pattern recombination for knowledge synthesis, and the parameter updates integrate optimization and task-specific knowledge across generations.

Algorithm 1 Evolutionary Synthesis Framework.

```

1: Initialize supercircuit  $S$  with architecture  $A$  and parameters  $\theta$ 
2: Define  $Fitness(c) = f(P(c), E(c))$  {Fitness function}
3: for generation  $g = 1$  to  $G$  do
4:    $\mathcal{P}_g \leftarrow$  Sample from  $S$ 
5:   for  $c \in \mathcal{P}_g$  do
6:     Evaluate  $P(c)$  {DAG path count}
7:     Evaluate  $E(c)$  {circuits expressivity}
8:   end for
9:    $\mathcal{P}'_g \leftarrow$  Select top  $K$  candidates based on  $Fitness(c)$ 
10:   $c_{best} \leftarrow \arg \max_{c \in \mathcal{P}'_g} Fitness(c)$ 
11:  for  $c_i \in \mathcal{P}'_g$  do
12:     $\theta^*_i \leftarrow \arg \min_{\theta} \mathcal{L}(c_i, \theta)$  {Trained focusing}
13:     $\theta^{new}_i \leftarrow \alpha \theta_{best} + (1 - \alpha) \theta^*_i$  {Parameter reuse}
14:  end for
15:  Update  $S$  with knowledge from  $c_{best}$ 
16:   $\mathcal{P}_{g+1} \leftarrow$  Evolve( $\mathcal{P}'_g$ ) {Crossover and mutation}
17:  Update  $A$  and  $\theta$  in  $S$  based on  $\mathcal{P}_{g+1}$ 
18: end for
19: return  $\arg \max_{c \in \mathcal{P}_G} Fitness(c)$ 

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4. Experiment results and performance analysis**4.1. Experiment framework**

A series of experiments were designed to validate the effectiveness of QUEST-A's knowledge-driven search strategy. These experiments include: signal representation tasks to demonstrate QUEST-A's performance in generative tasks, particularly in handling complex signals; image classification tasks to verify QUEST-A's performance in practical application scenarios; and variational quantum eigensolver (VQE) tasks to showcase its potential in quantum advantage domains. These experiments examine QUEST-A's performance across quantum systems of different scales and circuit architectures. Detailed information for experiments can be found in Appendix C.

4.2. Task I: QUEST-A in signal representation

The signal representation task aims to learn the intrinsic representation of signals, with its core objective being the learning of a function that maps input coordinates to corresponding signal values. In the field of quantum computing, this task demonstrates the capability of quantum models in continuous function approximation [33].

This experiment utilizes the same audio and image datasets as the benchmark models [24], ensuring consistency in data sources and preprocessing methods. Specifically, for the image representation task, images of astronaut, camera, and coffee from the scikit-image dataset [34] were selected.

For the dataset $D = (x_i, y_i)$, where x_i represents the coordinates of the signal y_i , the mean squared error (MSE) is utilized as the performance metric for the model output y'_i . The fitness function during the evolutionary process is defined as follows

$$Fitness := -\log \left(\frac{1}{|D|} \sum_{i=1}^{|D|} (y_i - y'_i)^2 \right). \quad (14)$$

The learning progression and final results of the model on the *Astronaut*, *Camera*, and *Coffee* images are visually represented in Fig. 5.

The results demonstrate the effectiveness of progressive knowledge accumulation in QUEST-A. In the initial stages of the QUEST-A method's search process, the candidate circuit population obtained through untrained filtering performed poorly, capturing almost no effective information. However, as knowledge accumulated through iterations, the performance of candidate circuits showed a rapid improvement trend. For the best-performing image, QUEST-A reduced the MSE from 1.1×10^{-3} of the baseline model to 0.45×10^{-3} , achieving approximately a 59.1% reduction in error rate. In the Astronaut image, which exhibits a more complex frequency pattern, QUEST-A similarly achieved a low error rate, demonstrating visually discernible advantages in capturing image details. Compared to the two baseline methods, QUEST-A exhibited superior performance in processing high-frequency information relative to the SOTA quantum methods, resulting in more precise retention of details and edge information in the generated images.

To further validate QUEST-A's capability in processing different types of signals, we also conducted experiments on audio representation tasks, with the results presented in Fig. 6.

The results of the audio representation task further confirm the advantages of the QUEST-A model in signal recognition. Although in terms of MSE, the QUEST-A model achieved only a 7.8% reduction in error rate compared to the baseline model, its performance in audio amplitude details processing surpassed state-of-the-art quantum methods. From the perspective of frequency graphs, the results generated by QUEST-A more accurately reproduced the frequency patterns of the benchmark audio, especially demonstrating more refined reproduction in the high-frequency components. While these details may contribute minimally to the overall MSE calculation, they are crucial for the perception of audio quality.

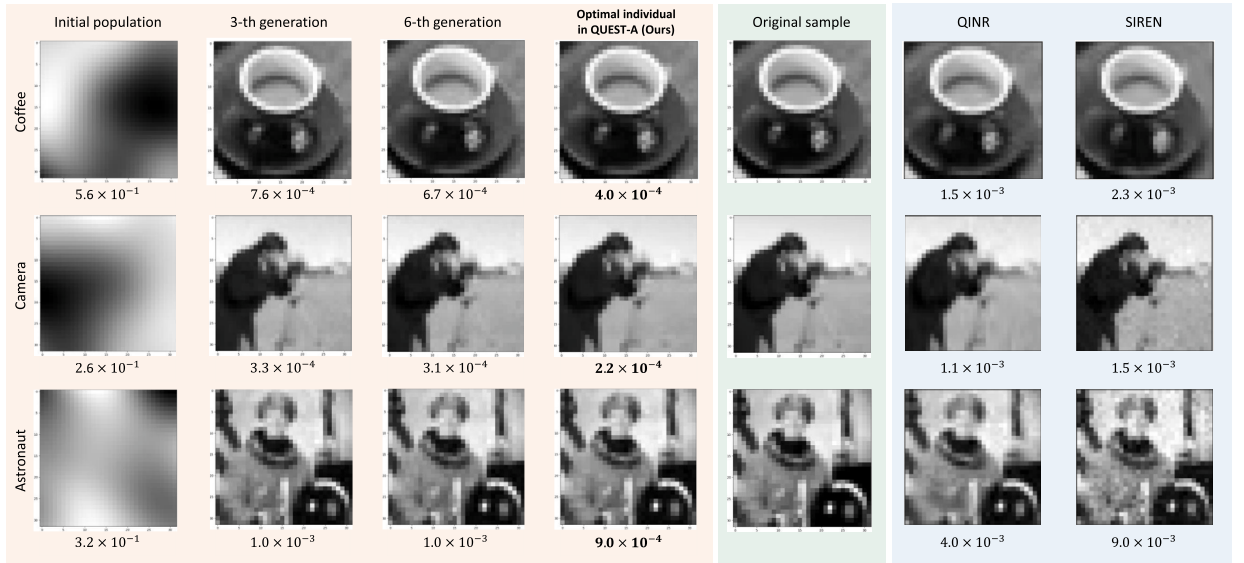


Fig. 5. Performance of QUEST-A in image representation tasks. The numerical values beneath each image represent the corresponding MSE. The red area illustrates the evolution of MSE for the best-performing individual in QUEST-A over generations. The green area shows the original images as the task benchmark. The blue area presents the results and MSE values for baseline [24,35].

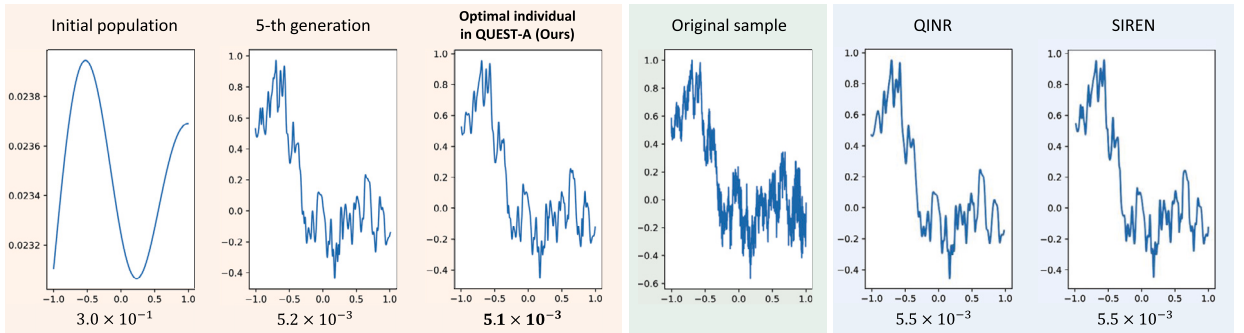


Fig. 6. Performance of QUEST-A in audio representation tasks. The red area illustrates the performance evolution of QUEST-A during the search process and its final results. The green area shows the original audio signals as the task benchmark. The blue area presents the performance of two baseline models on this task.

Table 1
Performance of QUEST-A and SOTA models in signal representation tasks.

Method	Sound Representation		Image Representation			
	Cello	params	Astronaut	Camera	Coffee	params ^a
SIREN	5.5 ^b	691	9.0	1.5	2.3	701
QIREN	5.5	649	4.0	1.1	1.5	657
QUEST-A (ours)	5.07(±0.06) (7.8%↑)	601 (7.4%↓)	1.83(±0.83) (54.1%↑)	0.45(±0.22) (59.1%↑)	0.65(±0.22) (56.3%↑)	609 (7.3%↓)

^a The params column indicates the number of parameters used by each model.

^b Performance is measured by the MSE($\times 10^{-3}$) between the representation results and the task benchmark, lower MSE values indicate better performance. (± standard deviation)

During the training process, we observed that the QUEST-A model quickly learned low-frequency patterns in the initial stages, similar to its performance in the image representation task. In the subsequent training process, the model made fine adjustments to the high-frequency components of the waveform and rapidly converged. This learning pattern further demonstrates QUEST-A's adaptability in handling different types of signal.

The comprehensive comparison with SOTA methods is presented in Table 1, showing superior performance across different signal types while maintaining lower parameter counts.

Table 2
Performance of QUEST-A and SOTA models in classification tasks.

Task	NNS ^a			SNS		RNE		
	Quilt	QNAS	QUEST-A (ours)	Élivágar ^b	QUEST-A (ours)	QNAS	Élivágar	QUEST-A (ours)
MNIST-2	94.5 ^c	95.0	99.9 ($\pm 0.3 \times 10^{-3}$)	97.0	99.28 (± 0.86)	95.0	93.0	96.5 (± 0.87)
MNIST-4	80.8	77.0	97.7 ($\pm 3.2 \times 10^{-3}$)	64.0	90.98 (± 0.81)	71.0	65.0	88.0 (± 1.70)
FMNIST-2	96.6	94.2	98.1 ($\pm 1.3 \times 10^{-3}$)	97.0	96.30 (± 0.40)	90.0	97.0	93.3 (± 0.79)
FMNIST-4	77.7	89.0	90.8 ($\pm 4.8 \times 10^{-3}$)	76.0	89.23 (± 2.92)	85.0	73.0	81.0 (± 1.64)

^a The table presents performance under three conditions: noiseless numerical simulation (NNS), simulated noisy simulation (SNS), and real noisy environment (RNE).

^b For baseline works, if multiple results were available under different conditions (e.g. different basic gate sets), the best performance was selected as the final performance for the baseline model.

^c Classification accuracy (%) were obtained through five independent experiments for QUEST-A.

Table 3
Performance of QUEST-A in multi-class classification tasks.

Dataset	Number of Classes ^a				
	2	4	6	8	10
MNIST	0.9994 ($\pm 0.3 \times 10^{-3}$)	0.9736 ($\pm 3.2 \times 10^{-3}$)	0.9391 ($\pm 3.5 \times 10^{-3}$)	0.9129 ($\pm 1.7 \times 10^{-3}$)	0.8545 ($\pm 2.0 \times 10^{-3}$)
FMNIST	0.9810 ($\pm 1.3 \times 10^{-3}$)	0.9087 ($\pm 4.8 \times 10^{-3}$)	0.8513 ($\pm 2.7 \times 10^{-3}$)	0.7576 ($\pm 3.5 \times 10^{-3}$)	0.7636 ($\pm 2.6 \times 10^{-3}$)

^a Performance of QUEST-A on MNIST and Fashion-MNIST for multi-class classification tasks.

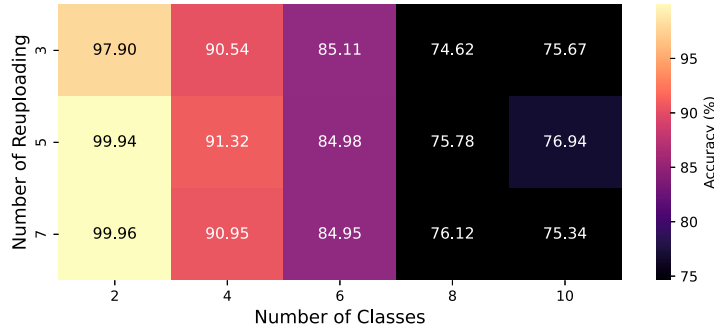


Fig. 7. Heatmap of Fashion-MNIST classification accuracy vs. number of re-uploading.

4.3. Task II: QUEST-A in image classification

Unlike signal representation tasks that focus on pattern discovery, image classification tasks emphasize the search for optimal decision boundaries. This study uses the widely adopted MNIST and Fashion-MNIST datasets as benchmarks, comparing QUEST-A with several state-of-the-art quantum baseline methods, including QUILT [36], Élivágar [1], and QNAS [19]. To thoroughly evaluate QUEST-A's performance in the NISQ era, we designed three experimental scenarios: numerical simulation, noise simulation, and experiments under real noisy environments. For the simulated noisy environment, we employed a composite noise model incorporating bit flip, phase flip, and depolarizing channel noise. The experimental results and performance comparisons on the MNIST and Fashion-MNIST datasets are presented in Table 2.

QUEST-A demonstrated superior search capability and robustness on both MNIST and Fashion-MNIST datasets. Compared to baseline models, QUEST-A achieved higher classification accuracy in noiseless environments (NNS). Notably, when noise was introduced (SNS), QUEST-A's performance showed only a slight decrease. This noise robustness became more evident as task difficulty increased; when transitioning from 2-class to 4-class classification, QUEST-A's performance degradation was less pronounced than that of baseline models. QUEST-A performed well on both MNIST and Fashion-MNIST datasets, indicating its generalization capability and adaptability. This robustness can be attributed to effective knowledge transfer between architectures. Further validation of QUEST-A's performance in Fashion-MNIST classification tasks is presented as Table 3 and Fig. 7.

In-depth analysis of image multi-classification tasks further validates the robustness of the QUEST-A model. QUEST-A demonstrated high performance stability across classification tasks ranging from 2 to 10 classes. Although model performance showed an

Table 4
Performance of QUEST-A and SOTA models in VQE tasks.

Systems	Qubit	PQAS	TF-QAS	QUEST-A (ours)
H_2	4	- ^a	1.6×10^{-8b}	$1.05(\pm 1.48) \times 10^{-11}$
Heisenberg	5	8.9×10^{-2}	1.2×10^{-3}	$6.07(\pm 10.6) \times 10^{-12}$
TFIM	6	9.0×10^{-2}	1.2×10^{-5}	$3.96(\pm 3.94) \times 10^{-7}$

^a The model consistently failed to complete the VQE task for H_2 in experimental reproduction.

^b Model performance is defined by the difference between the model's estimated value and the theoretical exact value.

expected declining trend as the number of classes increased, QUEST-A still maintained a high accuracy of over 75% in the most challenging 10-class classification task. Interestingly, increasing trainable layers does not necessarily lead to better performance, suggesting that our knowledge-guided search effectively identifies optimal architectures without over-parameterization.

Above experimental results not only further confirm QUEST-A's robustness in handling multi-classification tasks but also provide insights into the model's potential performance in more complex application scenarios.

4.4. Task III: QUEST-A in variational quantum eigensolver

The VQE task aims to estimate the ground state energy of a given Hamiltonian. In the quantum computing framework, this involves optimizing a parameterized quantum circuit to minimize the expected energy of the system. The VQE task primarily examines the model's ability to prepare complex quantum states, which should closely approximate the ground state of the target Hamiltonian, while also evaluating whether the ansatz design has sufficient expressive power to represent the solution space of the problem.

This experiment selects three typical systems as VQE objects: the H_2 molecule, the Heisenberg model, and the Transverse Field Ising Model. We selected two representative baseline models for comparison, TF-QAS [7] and PQAS [16], as fast architecture search method. Table 4. presents the performance of QUEST-A and these baseline models on three quantum systems along with detailed comparison results.

QUEST-A achieved the highest precision in all test cases, demonstrating an advantage of up to 11 orders of magnitude over state-of-the-art methods. Compared to similar fast architecture search methods, QUEST-A not only successfully completed the ground state energy prediction task within chemical accuracy ($1.6 \times 10^{-3} Ha$) but also achieved more precise predictions. As the system size increased from 4 qubits to 6 qubits, QUEST-A maintained its performance advantage. This outcome reflects QUEST-A's effectiveness in searching complex quantum architectures for ground state approximation. Its exceptional performance may stem from QUEST-A's ability to avoid getting trapped in local optima during the search process, as well as its enhanced sensitivity to the fine structure of quantum states.

The fact that this performance is achieved without relying on existing high-performance ansatzes further validates the effectiveness of our knowledge-driven search strategy in discovering optimal quantum circuit structures. This also demonstrates how the theoretical guarantees in search space reduction and knowledge transfer translate into practical performance improvements.

In conclusion, we evaluated our knowledge-driven framework across three representative quantum computing tasks. Our experimental results validate the theoretical insights about knowledge-guided search: in signal representation, QUEST-A achieved up to 59.1% performance improvement over SOTA methods through effective expressivity knowledge accumulation; in image classification, the framework maintained over 75% accuracy in challenging 10-class tasks across both simulated and noisy environments, demonstrating robust structural knowledge transfer; in VQE tasks, QUEST-A achieved precision improvements of up to 9 orders of magnitude through progressive knowledge refinement. These improvements support our theoretical analysis of how knowledge accumulation enables effective search space reduction and efficient architecture optimization, validating that systematic knowledge transfer translates into performance gains across diverse quantum computing scenarios.

5. Discussion

QUEST-A has been validated across diverse task complexities from standard benchmarks to complex applications, including image classification, VQE tasks, image similarity detection and medical image classification in Appendix F, demonstrating robustness and advantages at varying quantum system scales from 4 to 12 qubits.

The knowledge-driven framework proposed in this paper provides several insights into quantum architecture search. First, our theoretical analysis reveals that QAS can be effectively approached through a systematic knowledge accumulation process. Starting from limited prior understanding, the framework progressively discovers optimal architectures by enriching and transferring knowledge across different aspects of quantum circuits.

Second, our categorization of quantum circuit knowledge into structural, expressivity, optimization, and task-specific types proves to be practical. The structural and expressivity knowledge enable efficient space reduction, while optimization and task-specific knowledge guide precise architecture refinement. This multi-level knowledge representation naturally supports the balance between exploration and exploitation in architecture search.

Third, while our theoretical framework guarantees convergence under ideal conditions, practical applications may face additional challenges. For instance, the efficiency of knowledge transfer might be affected by the complexity of quantum circuits, and the

balance between different types of knowledge may need adjustment for specific tasks. On the other hand, while our framework includes implicit safeguards against negative knowledge transfer through parameter reuse rate control and fitness-based selection, the comprehensive impact of potentially harmful knowledge transfer between generations has not been thoroughly investigated in this work. Future research should explore more adaptive knowledge weighting mechanisms that can dynamically adjust knowledge transfer based on task similarity and architecture compatibility to further mitigate negative transfer risks.

Furthermore, our framework's success in diverse tasks suggests its potential extensibility to other quantum computing scenarios. The principles of progressive knowledge accumulation and multi-level knowledge transfer could be adapted for other quantum optimization problems, potentially offering new perspectives on quantum algorithm design and optimization.

6. Conclusion

In this work, we propose QUEST-A, a knowledge-driven framework for QAS through alternating phases of coarse-grained filtering and fine-grained focusing. This approach reformulates QAS as a knowledge acquisition and utilization process, decomposing the search process into progressive phases of knowledge accumulation. Experiments demonstrate QUEST-A's performance across signal representation, image classification, and VQE tasks, validating the effectiveness of our knowledge-driven approach.

Our exploration reveals three key insights into quantum architecture search. First, QAS can be seen as a knowledge accumulation process that progressively discovers optimal architectures through continuous enrichment of circuit understanding. Second, the multi-level knowledge categorization (structural, expressivity, optimization, and task-specific) provides a comprehensive framework for understanding and guiding the search process. Third, effective QAS requires dynamic knowledge-driven strategies rather than static search methods, where the process progressively learns and transfers architectural knowledge across different aspects of quantum circuits.

Future work should focus on developing adaptive knowledge-driven strategies in QAS, particularly exploring the design of quantum-principle-guided circuit templates as efficient search starting points and investigating task-specific knowledge transfer mechanisms.

CRedit authorship contribution statement

Lian-Hui Yu: Methodology, Data curation, Conceptualization. **Xiao-Yu Li:** Project administration, Methodology. **Geng Chen:** Writing – original draft, Investigation, Formal analysis. **Qin-Sheng Zhu:** Validation, Formal analysis. **Hui Li:** Software, Methodology. **Guo-Wu Yang:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Preliminaries of quantum circuit

The essence of quantum computing lies in embedding data into Hilbert space and processing information using quantum properties such as superposition and entanglement. Within this framework, quantum basic gates serve as fundamental units for manipulating quantum states.

In QAS task, basic gates are the fundamental units of circuit design. The core challenge lies in discovering optimal architectures through progressively understanding the characteristics and combinations of these basic gates. The basic gates used in this paper include R_X , R_Y , R_Z , and $CNOT$ gates, whose mathematical forms are as follows:

$$RX(\theta) = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, RY(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \quad (A.1)$$

$$RZ(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}, CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (A.2)$$

where, CNOT is a two-qubit gate, while the others act on single qubits. As derived in Section. 2.2, the challenge of QAS stems from the exponentially growing search space due to gate combinations and task-specific search objectives. This complexity motivates our knowledge-driven approach for efficient architecture discovery.

As discussed in Section 2.1, re-uploading circuits serve as an ideal template for our knowledge-driven QAS process by incorporating quantum encoding of classical data into the optimization scope. The basic architecture of a re-uploading circuit is illustrated in Fig. A.8.

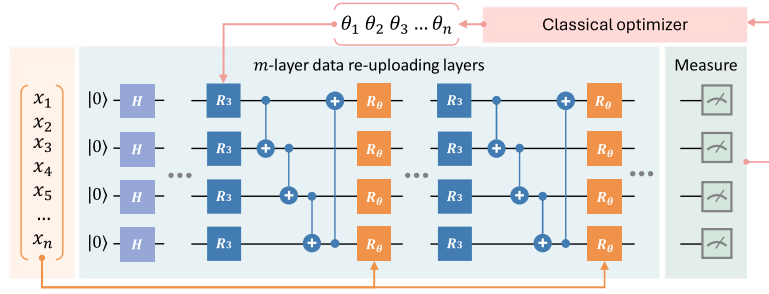


Fig. A.8. Learning framework based on re-uploading quantum circuits.

Appendix B. Theoretical performance analysis of knowledge-driven formulation

Based on this knowledge-driven formulation, we can theoretically analyze the efficiency and convergence properties of QAS. Our analysis focuses on how knowledge accumulation and transfer affect the search space complexity and optimization dynamics.

Theorem 1 (Knowledge-Space Complexity). *Given a knowledge state $\mathcal{K}$, the effective search space can be reduced to:*

$$S_{\text{eff}} = \{M \in S \mid f_s(M) \geq \alpha \wedge g_s(M) \in \text{Valid}\}, \quad (\text{B.1})$$

where $|S_{\text{eff}}| < |S|$.

Proof. Consider the original search space S for a quantum circuit with n qubits and m layers, where each position can select from k possible quantum gates. The total search space size is $|S| = O(k^{nm})$. The structural knowledge $\mathcal{K}_s = \{(f_s(M), g_s(M)) \mid M \in S\}$ provides two constraints:

$$S_1 = \{M \in S \mid f_s(M) \geq \alpha\}, \quad (\text{B.2})$$

which represents the subspace satisfying structural complexity requirements, and

$$S_2 = \{M \in S \mid g_s(M) \in \text{Valid}\}, \quad (\text{B.3})$$

which denotes the subspace meeting architectural feature requirements.

The effective search space is thus the intersection of these two subspaces:

$$S_{\text{eff}} = S_1 \cap S_2, \quad (\text{B.4})$$

where $|S_{\text{eff}}| \leq \min\{|S_1|, |S_2|\} \leq |S|$. Since these constraints are non-trivial, we have $|S_{\text{eff}}| < |S|$. $\square$

This proof demonstrates that through the application of structural knowledge, specifically the constraints on circuit complexity (f_s) and architectural features (g_s), the effective search space can be significantly reduced from the original space of size $O(k^{nm})$. The exact reduction magnitude depends on the specific knowledge utilized rather than hardware noise conditions or learning efficiency. This complexity reduction stems purely from the interplay between the original search space structure and the applied knowledge constraints, providing an upper bound that is independent of quantum-specific properties such as trainability or noise tolerance.

Theorem 2 (Knowledge Transfer Efficiency). *For consecutive knowledge states $\mathcal{K}_t$ and $\mathcal{K}_{t+1}$, the knowledge transfer complexity is:*

$$T(\mathcal{K}_t \rightarrow \mathcal{K}_{t+1}) = O(|C_t| + s + |D|)nm, \quad (\text{B.5})$$

where $|C_t|$ denotes the size of explored subspace at time t , s represents the sampling times, $|D|$ is the dataset size, and n, m are the number of qubits and layers respectively.

Proof. The knowledge transfer process involves updating four types of knowledge. Let us analyze their complexities respectively:

For structural knowledge $\mathcal{K}_s = \{(f_s(M), g_s(M)) \mid M \in S\}$, the update requires computing structural measures for all circuits in C_t . Each circuit analysis involves operations on n qubits through m layers, leading to complexity $T_s = O(|C_t|nm)$.

For expressivity knowledge $\mathcal{K}_e = \{(f_e(U), g_e(U)) \mid U = \sigma(M), M \in S\}$, the update involves s times of sampling and feature evaluation on quantum circuits of size nm , resulting in complexity $T_e = O(snm)$.

For optimization knowledge $\mathcal{K}_o = \{(\theta, \nabla L(\theta), \eta(\theta)) \mid \theta \text{ are parameters of circuits in } C\}$, the update involves parameter gradient computation for current circuit with size nm , yielding complexity $T_o = O(nm)$.

For task-specific knowledge $\mathcal{K}_t = \{(M, L(U, D), v(M)) \mid M \in C\}$, the update requires loss computation on dataset D for circuits of size nm , leading to complexity $T_t = O(|D|nm)$.

Since all four types of knowledge need to be updated sequentially in each transfer process, the total complexity is the sum of individual complexities:

$$T(\mathcal{K}_t \rightarrow \mathcal{K}_{t+1}) = O((|C_t| + s + |D|)nm) \quad \square \quad (\text{B.6})$$

This complexity analysis provides insights into the efficiency trade-offs between different knowledge utilization strategies in QAS. Consider three scenarios:

1. For QAS without knowledge guidance, the process essentially becomes random search in the original space $O(k^{nm})$, with complexity $O(k^{nm} \cdot |D|nm)$ where only basic task evaluation is needed for each circuit.
2. When partial knowledge is utilized, e.g., only structural knowledge, the per-iteration complexity reduces to $O(|C_t|nm)$. The search space can be reduced to $O(k^{nm-\alpha})$, where α represents the reduction effect from partial knowledge constraints. The overall complexity becomes $O(k^{nm-\alpha} \cdot |C_t|nm)$.
3. With complete knowledge utilization, although the per-iteration complexity increases to $O((|C_t| + s + |D|)nm)$ due to comprehensive knowledge transfer, the search space can be further reduced to $O(k^{nm-\beta})$ where $\beta > \alpha$. This leads to an overall complexity of $O(k^{nm-\beta} \cdot (|C_t| + s + |D|)nm)$.

Notably, there exists a multi-faceted trade-off in knowledge utilization. While incorporating more knowledge types increases the per-iteration computational cost as $O((|C_t| + s + |D|)nm)$, it enables more effective search space reduction from $O(k^{nm})$ to $O(k^{nm-\beta})$.

The efficiency of knowledge utilization correlates with search space scale. For small search spaces where $nm \cdot \log k$ is relatively small (e.g., few-qubit shallow circuits), the overhead of complete knowledge transfer dominates the computational cost, making partial knowledge utilization more efficient. For large search spaces where $nm \cdot \log k$ is substantial (e.g., deep circuits with many qubits), the reduction effect from comprehensive knowledge utilization becomes dominant, as the decreased number of circuit evaluations outweighs the knowledge transfer overhead of $O((|C_t| + s + |D|)nm)$.

In noisy quantum environments, the trade-off between knowledge transfer overhead and search efficiency manifests distinctly. Noise introduces additional computational overhead in circuit evaluation, as each circuit requires multiple executions to mitigate noise effects. This evaluation overhead scales with the search space size: larger spaces require more circuit evaluations under noise. Knowledge-guided search, despite its higher per-iteration complexity of $O((|C_t| + s + |D|)nm)$, achieves efficiency through two mechanisms: first, it reduces the total number of circuits requiring noise-affected evaluation by constraining the search space to $O(k^{nm-\beta})$; second, it enables the identification of noise-resilient circuit structures through architectural constraints derived from structural knowledge. Thus, in noisy environments, the computational advantage of knowledge utilization becomes more pronounced.

This analysis establishes that the efficiency of knowledge-guided QAS depends on quantum noise levels, search space dimensionality, and computational constraints, necessitating balanced knowledge utilization strategies based on specific search contexts.

Theorem 3 (Convergence). *Under the knowledge-driven framework with proper knowledge accumulation and transfer, QAS process converges.*

Proof. The convergence can be established through the following aspects:

First, from Theorem 1, the search process is conducted in a finite space S_{eff} where $|S_{\text{eff}}| < |S|$. Denote $\mathcal{U}_t$ as the unexplored space at iteration t , where $\mathcal{U}_0 = S_{\text{eff}}$. Second, from Theorem 2, for any two consecutive knowledge states $\mathcal{K}_t$ and $\mathcal{K}_{t+1}$, the knowledge transfer has bounded complexity $T(\mathcal{K}_t \rightarrow \mathcal{K}_{t+1}) = O((|C_t| + s + |D|)nm)$. This ensures the effectiveness of each iteration.

For iteration t , denote M_t^* as the best architecture found up to iteration t . To measure the potential improvement in current unexplored space, we define the improvement metric at iteration t as:

$$\Delta_t = \min_{M \in \mathcal{U}_t} L(M) - L(M_t^*), \quad (\text{B.7})$$

where $L(\cdot)$ represents the loss function. This metric indicates whether better architectures exist in the unexplored space $\mathcal{U}_t$. When $\Delta_t \geq 0$, no better architecture exists in current unexplored space; when $\Delta_t < 0$, there exists at least one architecture with better performance than M_t^* .

Based on this metric, the unexplored space is updated after each iteration:

$$\mathcal{U}_{t+1} = \mathcal{U}_t \setminus \{M \mid L(M) \geq L(M_t^*)\}. \quad (\text{B.8})$$

This equation describes the search space reduction mechanism: in each iteration, all architectures with performance no better than the current optimal architecture M_t^* are removed from the unexplored space. This updating mechanism ensures the unexplored space strictly decreases as $\mathcal{U}_{t+1} \subset \mathcal{U}_t$, and each update is effective as $|\mathcal{U}_{t+1}| < |\mathcal{U}_t|$ when $\mathcal{U}_t \neq \emptyset$.

Given that $\mathcal{U}_0$ is finite from Theorem 1, and the size of unexplored space strictly decreases in each iteration ($|\mathcal{U}_{t+1}| < |\mathcal{U}_t|$ when $\mathcal{U}_t \neq \emptyset$), there must exist a finite T where $\mathcal{U}_T = \emptyset$, indicating the completion of architecture exploration in the effective search space S_{eff} .

Therefore, the knowledge-driven QAS process converges in finite steps. $\square$

The above theoretical analysis provides key insights into why and how knowledge-driven QAS can achieve effective performance. The Knowledge-Space Complexity theorem demonstrates that through proper knowledge constraints, we can reduce the search space

Table C.5
Detailed parameter settings for all experiments in our work.

Parameter	Signal Representation	Image Classification	VQE
Number of qubits	8	10	5 (<i>Heisenberg</i>) 4 (<i>H₂</i>), 6 (<i>TFIM</i>)
Data re-uploading	2	3	- ^a
Number of generations	10	10	10
Population size	5	5	5
Loss metric	MSE	Crossentropy	-
Trainable parameters ^b	601 (<i>Sound</i>) 609 (<i>Image</i>)	175	175 (<i>Heisenberg</i>) 105 (<i>H₂</i>), 420 (<i>TFIM</i>)

^a The VQE task does not utilize specific circuit templates, nor does it employ gradient-based optimization processes.

^b Experimental parameters are set to be comparable with baseline models.

from the original size $|S|$ to a smaller effective space $|S_{\text{eff}}|$, where $|S_{\text{eff}}| < |S|$. The Knowledge Transfer Efficiency theorem establishes a bounded complexity of $O((|C_t| + s + |D|)nm)$ for knowledge transfer between consecutive states, revealing a trade-off between knowledge utilization and computational cost. This trade-off becomes particularly significant in different contexts such as noisy environments and varying search space scales. The Convergence theorem further proves that this knowledge-driven approach guarantees convergence through a systematic space reduction mechanism, where the unexplored space strictly decreases in each iteration until the optimal architecture is found.

Appendix C. Experimental details

Numerical simulation experiments were conducted on a Python-based simulation platform, utilizing PennyLane, Mindspore Quantum [37], and TensorFlow Quantum. Experiments in a real quantum computing environment were performed on the *ibm_kyiv* backend. The [code](#) for QUEST-A is available.

In the search process, the best-performing individual is selected as the final performance metric for the experiment. All experiments were conducted independently five times, average performance with standard deviation reported as the final result. The experimental settings for our work are presented in the Table C.5, and detailed process for each task are as follows:

Task I: QUEST-A in Signal Representation

Images for the task were first downsampled to a resolution of 32×32 pixels and then resized through a linear layer before being input into a 10-qubit quantum model. The model processed the data through three iterations of data re-uploading for architecture discovery. Additionally, the model is equipped with two variational layers, structurally identical to the re-uploading circuits, both at the initial stage and prior to the measurement stage. These layers serve to optimize the initial state of quantum evolution and the basis vectors for measurement. The evolutionary algorithm was configured to run for 10 generations, with each generation comprising 10 individuals.

Task II: QUEST-A in Image Classification

In terms of preprocessing, all 28×28 pixel images first undergo principal component analysis (PCA), after which the processed data is input into a 10-qubit quantum circuit. The model performs four iterations of data re-uploading and uses cross-entropy as the loss function.

Task III: QUEST-A in VQE

This experiment selects three typical systems as VQE objects: the H_2 molecule (4 qubits), the Heisenberg model (5 qubits), and the Transverse Field Ising Model (TFIM, 6 qubits). Due to the unique characteristics of the VQE, QUEST does not employ data re-uploading circuits in this task. The gate set used for the task includes RX , RY , RZ , and $CNOT$ gates. The Hamiltonian for the H_2 molecule follows the construction method from previous work [38]. The Hamiltonians for TFIM and the Heisenberg model are defined as follows:

$$H_{TFIM} = \sum_{i=1}^n Z_i Z_{i+1} - X_i, \quad (C.1)$$

$$H_{Heis} = \sum_{i=1}^n X_i X_{i+1} + Y_i Y_{i+1} + Z_i Z_{i+1} + \sum_{i=1}^n z_i, \quad (C.2)$$

where n denotes the number of qubits. The model's performance is measured by the difference between the estimated value E and the theoretically exact ground state energy E_0 , i.e., $E - E_0$. To ensure fair comparison, QUEST-A and the baseline models use the same number of qubits and similar parameter counts. In each experiment, the models are run multiple times, with the best performance selected as the final result.

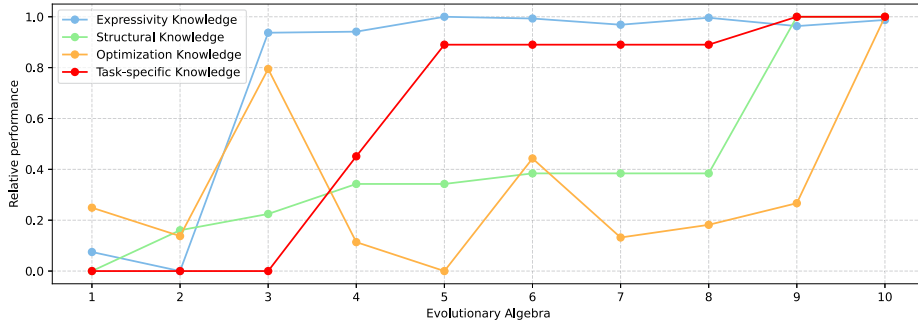
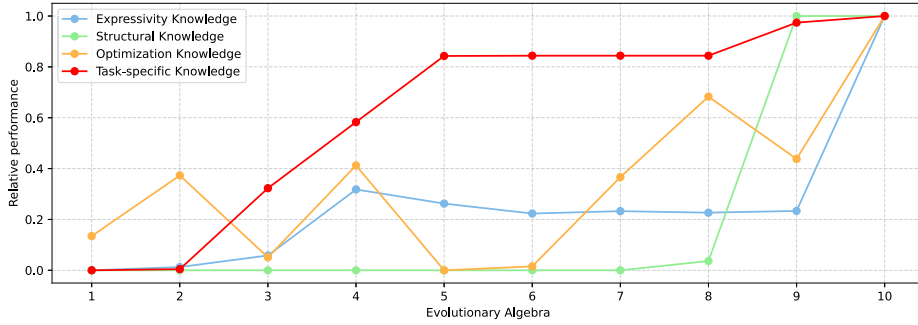
Fig. D.9. Knowledge tracing for image representation tasks of *Coffee*.

Fig. D.10. Knowledge tracing for audio representation tasks.

Appendix D. Analysis of knowledge evolution in QAS

To validate our theoretical framework of knowledge-driven QAS, we conduct a quantitative analysis of knowledge evolution during the search process. This analysis serves two purposes: (1) providing empirical evidence for our theoretical claims about knowledge accumulation and transfer, and (2) offering insights into the dynamics of different knowledge types in guiding architecture search.

To quantitatively analyze the evolution of different knowledge types, we establish the following metrics:

- **Structural Knowledge** is quantified by the number of edges in the DAG representation of quantum circuits, as established in Equation (3). This metric directly reflects the structural complexity and connectivity patterns of quantum architectures.
- **Expressivity Knowledge** is measured through KL divergence between the circuit's output distribution and Haar random distribution, following Equation (4). This measure captures the circuit's capability in exploring quantum state space.
- **Optimization Knowledge** is evaluated using the Convergence Index, defined as the loss value when reaching 80% convergence during training. This metric aligns with the optimization dynamics described in Equation (5).
- **Task-specific Knowledge** is quantified by the fitness function value as defined in Equation (6), which directly measures how well the architecture performs on the target task.

These metrics are averaged over five independent experiments and normalized to $[0, 1]$ scale, where values approaching 0 indicate initial knowledge states and values approaching 1 represent mature knowledge accumulation. This normalization enables direct comparison of knowledge evolution trends across different types, with higher values reflecting more comprehensive knowledge acquisition in their respective aspects.

For the coffee image representation task in Fig. D.9, we observe distinct evolution patterns across different knowledge types. The expressivity knowledge, measured by KL divergence, rapidly increases in the second generation and stabilizes thereafter, indicating that basic representation capability is quickly established. Structural knowledge shows steady improvement with a sharp increase in later generations, suggesting progressive refinement of circuit architecture. The optimization knowledge exhibits oscillatory behavior before final convergence, reflecting the complex dynamics of parameter optimization. The task-specific fitness demonstrates rapid improvement between generations 3 ~ 5, coinciding with the maturation of other knowledge types. These patterns align with our theoretical framework: the quick stabilization of expressivity knowledge supports our hypothesis about efficient knowledge-guided filtering, while the steady improvement in structural knowledge validates the effectiveness of our evolutionary knowledge transfer mechanism. The final convergence of all metrics suggests successful knowledge synthesis as predicted by Theorem 3.

For the audio representation task in Fig. D.10, we observe a distinct performance plateau between generations 5-7. During this period, task-specific knowledge remains relatively stable while structural knowledge and optimization knowledge exhibit exploratory fluctuations. Following this brief plateau, accompanied by a breakthrough in expressivity knowledge (generation 8), task performance

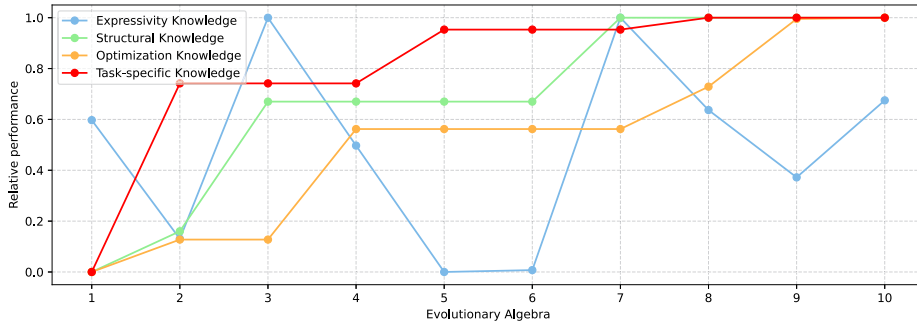
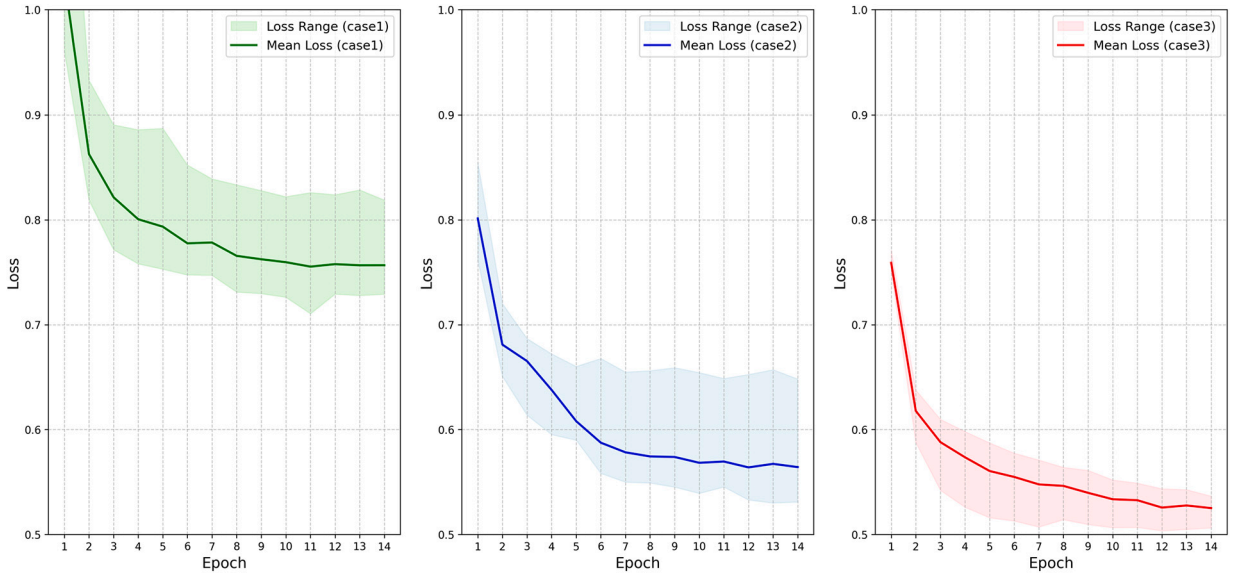
Fig. D.11. Knowledge tracing for VQE tasks of H_2 .

Fig. E.12. Loss trend over epochs for three case. Training progress shown until convergence of case i.

achieves significant improvement again. This phenomenon of escaping local optima validates the effectiveness of our knowledge-driven search strategy.

For the VQE task of H_2 in Fig. D.11, the fitness value exhibits rapid initial growth followed by step-wise improvements. The structural knowledge shows step-wise increases in both early and late generations, while expressivity knowledge remains stable. Although the expressivity knowledge appears to fluctuate significantly in the visualization due to the relative scaling of the normalized metrics, its absolute numerical changes are minimal, reflecting the relatively simple circuit structures in VQE tasks. In contrast to generative tasks, this distinct evolution pattern demonstrates our framework's adaptability across different quantum computing scenarios.

The knowledge tracing analysis across different tasks reveals distinct evolution patterns among knowledge types, suggesting potential independence in their exploration processes. The observed variations, particularly the contrasting behavior in expressivity knowledge between generative and VQE tasks, provide insights into task-specific optimization strategies. These findings suggest potential directions for improving QAS methods, such as task-adaptive knowledge prioritization and targeted refinement of VQE methodologies based on knowledge evolution patterns.

Appendix E. Ablation analysis

We conducted a set of ablation experiments to validate the effectiveness of progressive knowledge learning using the Fashion-MNIST 10-class classification task as the experimental scenario. The experiments investigated the following three cases: (i) Trained focusing in random circuits. (ii) Trained focusing after DAG-based untrained filtering only; (iii) Trained focusing after DAG and KL-based untrained filtering;

For random trained focusing (case i), we used 10 randomly generated circuits for focusing. For the untrained filtering cases (cases ii and iii), we screened from 100 candidate circuits and selected the top 10 optimal circuits. All experiments were terminated upon convergence of the loss function. See Fig. E.12.

Table F.6
Performance of QUEST-A and baseline in image similarity detection tasks.

Dataset	SliQ	QUSL	QUEST-A(ours)
CIFAR_10	0.706(± 0.052)	0.790(± 0.059)	0.894 (± 0.092)
ImagNet	0.695(± 0.042)	0.813(± 0.030)	0.878 (± 0.117)

Case (iii) with complete untrained filtering exhibited faster convergence and superior performance compared to the other two cases. Case (i) with random circuits demonstrated slower convergence and inferior final performance. Comparing case (i) with cases using untrained filtering, it is evident that the filtering process effectively enhanced the initial search space and convergence capability, indicating better architecture discovery potential. Moreover, the comprehensive untrained filtering process in case (iii) versus case (ii) yielded further improvements in both convergence speed and final performance.

Appendix F. Applications of QUEST-A in real-world scenarios

To validate the effectiveness of QUEST-A in practical scenarios, we apply our method to two challenging real-world tasks: image similarity detection [39] and medical image classification [40].

F.1. Image similarity detection

The image similarity task can be viewed as an infinite-class classification problem, serving as a foundational component for complex image processing applications such as object detection and recognition. We employ a 12-qubit quantum circuit with three re-uploading layers. The circuit architecture follows standard template but with increased qubit capacity to handle the more complex feature representations required for similarity computations. This task remains relatively unexplored in quantum computing settings, with only preliminary investigations. We compare QUEST against pioneering works SliQ [39] and representative QAS-based approaches QUSL [12], maintaining consistency with their image preprocessing and feature extraction procedures. The performance is evaluated using the Spearman correlation coefficient between the similarity detection results and the Euclidean distances of image pairs, experiments are performed on CIFAR_10 [41] and ImagNet [42] in Table F.6.

QUEST surpassing both the pioneering SliQ method and the QAS-based QUSL approach in task performance. These results indicate that knowledge-driven framework can effectively capture image similarities in complex feature spaces, demonstrating its applicability to challenging computer vision tasks.

F.2. Medical image classification

The medical image classification experiments utilize three representative datasets from MedMNIST: PneumoniaMNIST (binary classification of Chest X-Rays), RetinaMNIST (ordinal regression of Fundus Camera images), and BreastMNIST (binary classification of Breast Ultrasound). We implement a 10-qubit circuit with three re-uploading layers, evaluating performance using both Area Under Curve (AUC) and accuracy (ACC) metrics. To provide comprehensive validation, we compare QUEST-A against four categories of baselines:

- Hand-crafted quantum neural networks: Quantum Medical Image Classification (QMIC) [43], Quantum Vision Transformers (QVT) [44].
- Classical neural architecture search methods: NSGA-NET [45], HOPNAS [46], SNAS [47], DARTS [48], and AutoKeras [49].
- Commercial vision platforms: Google AutoML Vision (GAMV) [40].
- Standard deep learning models: ResNet-50(28) [40] and auto-sklearn [50].

The experimental results across these datasets and baseline methods are presented in Table F.7.

The results demonstrate competitive performance of QUEST-A in medical image classification tasks. Compared with specialized quantum methods specifically designed for medical image classification like QMIC and QVT, our approach achieves comparable results on PneumoniaMNIST and BreastMNIST. When compared with mainstream classical methods, QUEST-A maintains reasonable performance despite using smaller model scales. These results validate our method's adaptability across different quantum computing scenarios while suggesting potential for improvement. Future enhancements could focus on incorporating domain-specific quantum compression techniques, adjusting knowledge weightings for medical imaging tasks, and developing specialized ansatz designs for healthcare applications.

Data availability

All data generated or analyzed during this study are available and included in this published article. [Code](#) for our work has been open-sourced.

Table F.7

Performance of QUEST-A and baseline in medical image datasets.

Methods	PneumoniaMNIST		RetinaMNIST		BreastMNIST	
	AUC	ACC	AUC	ACC	AUC	ACC
NSGA-NET	0.965	0.907	0.759	0.540	0.857	0.846
GAMV	0.991	0.946	0.750	0.531	0.919	0.861
ResNet-50(28)	0.948	0.854	0.726	0.528	0.857	0.812
HOPNAS	0.971	0.852	0.770	0.523	0.907	0.853
auto-sklearn	0.942	0.855	0.690	0.515	0.836	0.803
SNAS	0.974	0.871	0.753	0.515	0.894	0.811
DARTS	0.965	0.874	0.742	0.510	0.912	0.832
AutoKeras	0.947	0.878	0.719	0.503	0.871	0.831
QMIC	0.900	0.830	0.840	0.800	-	-
QVT	0.947	0.885	0.745	0.542	0.770	0.744
QUEST-A	0.912	0.838	0.690	0.526	0.781	0.801
(ours)	(±0.003)	(±0.011)	(±0.010)	(±0.016)	(±0.001)	(±0.011)

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